

9.1 LU -Decompositions

Up to now, we have focused on Gaussian elimination and Gauss–Jordan elimination for solving linear systems. While suitable for small problems, these methods are not ideal for large-scale problems where computer roundoff error, memory, and speed are concerns. In this section we discuss a method based on factoring the coefficient matrix into a product of lower and upper triangular matrices—the LU -decomposition—which underpins many computer algorithms.

Solving Linear Systems by Factoring

Our first goal is to show how to solve $A\mathbf{x} = \mathbf{b}$ (an $n \times n$ system) by factoring the coefficient matrix A .

Definition 1

A factorization of a square matrix A as

$$A = LU \quad (1)$$

where L is lower triangular and U is upper triangular is called an **LU -decomposition** (or **LU -factorization**) of A .

The Method of LU -Decomposition

Step 1. Rewrite $A\mathbf{x} = \mathbf{b}$ as

$$LU\mathbf{x} = \mathbf{b} \quad (2)$$

Step 2. Substitute $\mathbf{y} = U\mathbf{x}$ and rewrite (2) as $L\mathbf{y} = \mathbf{b}$; solve for $\mathbf{y}$ (*forward substitution*).

Step 3. Substitute $\mathbf{y}$ into $U\mathbf{x} = \mathbf{y}$ and solve for $\mathbf{x}$ (*back substitution*).

This procedure replaces the single system $A\mathbf{x} = \mathbf{b}$ by the two triangular systems $L\mathbf{y} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$, which are solved in succession. Because each system has a triangular coefficient matrix, the total computation is generally no more than solving the original system directly.

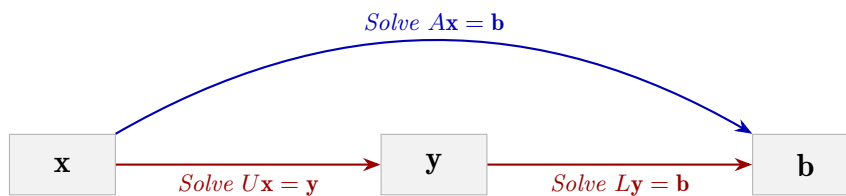


FIGURE 9.1.1

EXAMPLE 1 | Solving $A\mathbf{x} = \mathbf{b}$ by LU -Decomposition

Use the LU -decomposition

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}}_U \quad (4)$$

to solve the system $\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$.

SOLUTION: Rewriting as $LU\mathbf{x} = \mathbf{b}$ and setting $\mathbf{y} = U\mathbf{x}$, we first solve $L\mathbf{y} = \mathbf{b}$:

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \implies \begin{aligned} 2y_1 &= 2 \\ -3y_1 + y_2 &= 2 \\ 4y_1 - 3y_2 + 7y_3 &= 3 \end{aligned}$$

Forward substitution gives $y_1 = 1$, $y_2 = 5$, $y_3 = 2$. Substituting into $U\mathbf{x} = \mathbf{y}$:

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} \implies \begin{aligned} x_1 + 3x_2 + x_3 &= 1 \\ x_2 + 3x_3 &= 5 \\ x_3 &= 2 \end{aligned}$$

Back substitution yields $x_1 = 2$, $x_2 = -1$, $x_3 = 2$.

Finding LU -Decompositions

Once an LU -decomposition of A is found, solving a sequence of systems $A\mathbf{x} = \mathbf{b}_1$, $A\mathbf{x} = \mathbf{b}_2$, $\dots$, $A\mathbf{x} = \mathbf{b}_k$ requires factoring A only once; the factorization is then reused for each right-hand side. This is the main advantage of LU -decomposition.

Not every square matrix has an LU -decomposition. However, if A can be reduced to row echelon form by Gaussian elimination *without row interchanges*, then it has one. Specifically, if elementary matrices $E_1, E_2, \dots, E_k$ (none involving row interchanges) reduce A to U , so that

$$E_k \cdots E_2 E_1 A = U \quad (8)$$

then $A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU$, where

$$L = E_1^{-1} E_2^{-1} \cdots E_k^{-1} \quad (10)$$

Theorem 9.1.1

If A is a square matrix that can be reduced to row echelon form U by Gaussian elimination without row interchanges, then A can be factored as $A = LU$, where L is lower triangular.

Proof. Let L and U be as in Formulas (10) and (8). U is upper triangular because it is a row echelon form of a square matrix. Each E_j (no row interchange) either adds a scalar multiple of one row to a lower

row, or scales a row by a nonzero constant; in both cases E_j is lower triangular, and by Theorem 1.7.1(d) so is E_j^{-1} . Hence L , a product of lower triangular matrices, is lower triangular. ■

EXAMPLE 2 | An LU -Decomposition

Find an LU -decomposition of $A = \begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$.

SOLUTION: We reduce A to row echelon form without row interchanges and track the elementary matrices and their inverses.

Reduction step	Row op	E_j	E_j^{-1}
Step 1: $\frac{1}{2} \times \text{row 1}$	$\frac{1}{2}r_1$	$\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Step 2: $3r_1 + r_2$	$(3 \times r_1) + r_2$	$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Step 3: $-4r_1 + r_3$	$(-4 \times r_1) + r_3$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$
Step 4: $3r_2 + r_3$	$(3 \times r_2) + r_3$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$
Step 5: $\frac{1}{7} \times \text{row 3}$	$\frac{1}{7}r_3$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{7} \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$

The result is $U = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$, and computing $L = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}$ yields

$$L = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \quad (11)$$

so $A = LU$ agrees with the factorization in Example 1.

Bookkeeping

The matrix computation in Example 2 is unnecessary. Observe:

- The **diagonal entries** of L are the reciprocals of the multipliers used to introduce leading 1's in U .
- Each **off-diagonal entry** of L below the main diagonal is the *negative* of the multiplier used to introduce the zero in that position in U .

This gives a streamlined 4-step procedure:

Procedure for Constructing an LU -Decomposition

Step 1. Reduce A to row echelon form U by Gaussian elimination without row interchanges, tracking the multipliers for each leading 1 and each zero introduced below a leading 1.

Step 2. Along the main diagonal of L , place the *reciprocal* of each multiplier that introduced a leading 1 in U .

Step 3. At each position below the main diagonal of L , place the *negative* of the multiplier used to introduce the zero at that position in U .

Step 4. Form the decomposition $A = LU$.

EXAMPLE 3 | Constructing an LU -Decomposition

Find an LU -decomposition of $A = \begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}$.

SOLUTION: We reduce A to U while filling in L simultaneously (bullets denote not-yet-determined entries of L).

$$A = \begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}, \quad L = \begin{bmatrix} \bullet & 0 & 0 \\ \bullet & \bullet & 0 \\ \bullet & \bullet & \bullet \end{bmatrix}$$

Step 1: $\frac{1}{6} \times r_1$ (multiplier = $\frac{1}{6}$, so $l_{11} = 6$):

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}, \quad L_{11} = 6$$

Step 2: $(-9) \times r_1 + r_2$ and $(-3) \times r_1 + r_3$ (multipliers -9 and -3 , so $l_{21} = 9$, $l_{31} = 3$):

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 2 & 1 \\ 0 & 8 & 5 \end{bmatrix}, \quad L = \begin{bmatrix} 6 & 0 & 0 \\ 9 & \bullet & 0 \\ 3 & \bullet & \bullet \end{bmatrix}$$

Step 3: $\frac{1}{2} \times r_2$ (multiplier = $\frac{1}{2}$, so $l_{22} = 2$):

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 8 & 5 \end{bmatrix}, \quad L_{22} = 2$$

Step 4: $(-8) \times r_2 + r_3$ (multiplier -8 , so $l_{32} = 8$):

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & \bullet \end{bmatrix}$$

Step 5: r_3 already has leading 1 (multiplier = 1, so $l_{33} = 1$):

$$U = \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix}$$

Thus the LU -decomposition is

$$\begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

LU -Decompositions Are Not Unique

In general, LU -decompositions are not unique. If $A = LU$ and L has nonzero diagonal entries (guaranteed when A is invertible), we can shift diagonal entries from L to a separate diagonal matrix D , writing $L = L'D$ where L' has 1's on the main diagonal. This gives another LU -decomposition $A = (L')(DU)$.

LDU -Decompositions

If it is preferred to have 1's on the main diagonal of *both* the lower and upper triangular factors, we can factor as $L = L'D$ and write

$$A = L'DU$$

where L' has 1's on the diagonal, D is diagonal, and U has 1's on the diagonal. This is the **LDU -decomposition** of A . For example,

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad (13)$$

If A is invertible and can be reduced to row echelon form without row interchanges, then A can be factored *uniquely* as $A = LDU$.

Exercise Set 9.1

In Exercises 1–2, use the given LU -decomposition to solve the system.

$$1. \quad \begin{bmatrix} 3 & -6 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}; \quad \text{solve } 3x_1 - 6x_2 = 0, \quad -2x_1 + 5x_2 = 1.$$

$$2. \quad \begin{bmatrix} 3 & -6 & -3 \\ 2 & 0 & 6 \\ -4 & 7 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 4 & 0 \\ -4 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}; \quad \text{solve } 3x_1 - 6x_2 - 3x_3 = -3, \quad 2x_1 + 6x_3 = -22, \\ -4x_1 + 7x_2 + 4x_3 = 3.$$

In Exercises 3–6, find an LU -decomposition of the coefficient matrix and then use the method of Example 1 to solve the system.

$$3. \begin{bmatrix} 2 & 8 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$5. \begin{bmatrix} 2 & -2 & -2 \\ 0 & -2 & 2 \\ -1 & 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ 6 \end{bmatrix}$$

In Exercises 7–8, an LU -decomposition of A is given.

a. Compute L^{-1} and U^{-1} .

b. Use the result in part (a) to find the inverse of A .

$$7. A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 1 \\ -6 & -1 & 2 \end{bmatrix}; \quad A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & -5 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\text{Exercise 9. Let } A = \begin{bmatrix} 2 & 1 & -1 \\ -2 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}.$$

a. Find an LU -decomposition of A .

b. Express A in the form $A = L_1 D U_1$, where L_1 is lower triangular with 1's along the main diagonal, U_1 is upper triangular with 1's along the main diagonal, and D is a diagonal matrix.

c. Express A in the form $A = L_2 U_2$, where L_2 is lower triangular with 1's along the main diagonal and U_2 is upper triangular.

$$10. \text{ a. Show that the matrix } \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ has no } LU\text{-decomposition.}$$

b. Find a PLU -decomposition of this matrix.

In Exercises 11–12, use the given PLU -decomposition of A to solve $A\mathbf{x} = \mathbf{b}$ by rewriting it as $P^{-1}A\mathbf{x} = P^{-1}\mathbf{b}$ and solving by LU -decomposition.

$$12. \mathbf{b} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}; \quad A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 2 & 1 \\ 8 & 1 & 8 \end{bmatrix}; \quad A = \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_P \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{4} & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 8 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{9}{4} \end{bmatrix}}_U = PLU$$

In Exercises 13–14, find the LDU -decomposition of A .

$$13. A = \begin{bmatrix} 2 & 2 \\ 4 & 1 \end{bmatrix}$$

$$14. A = \begin{bmatrix} 3 & -12 & 6 \\ 0 & 2 & 0 \\ 6 & -28 & 13 \end{bmatrix}$$

In Exercises 15–16, find a PLU -decomposition of A , and use it to solve $A\mathbf{x} = \mathbf{b}$.

$$16. \ A = \begin{bmatrix} 0 & 3 & -2 \\ 1 & 1 & 4 \\ 2 & 2 & 5 \end{bmatrix}; \quad \mathbf{b} = \begin{bmatrix} 7 \\ 5 \\ -2 \end{bmatrix}$$

Working with Proofs

17. Let $A\mathbf{x} = \mathbf{b}$ be a linear system of n equations in n unknowns, and assume A is invertible and can be reduced to row echelon form without row interchanges. How many additions and multiplications are required to solve the system by the method of Example 1?